

KIRKLAND LAKE, ON, Canada

WMO: 713010

Lat: 48.151N

Lon: 80.012W

Elev: 324

StdP: 97.49

Time Zone: -5.00 (NAE)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -33.8	(c) -30.4	(d) -36.4	(e) 0.1	(f) -33.7	(g) -32.9	(h) 0.2	(i) -30.1	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	29.0	20.0	27.1	18.8	25.3	18.0	21.9	26.1	20.8	24.2	19.7	23.0	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.6	15.9	23.5	19.6	14.9	22.4	18.6	14.0	21.7	65.6	26.4	61.7	24.0	58.0	23.0	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) N/A	(b) N/A	(c) N/A	(e) -38.5	(f) 31.9	(g) 3.2	(h) 1.7	(i) -40.8	(j) 33.1	(k) -42.7	(l) 34.1	(m) -44.5	(n) 35.0	(o) -46.8	(p) 36.3		
(4) N/A	(5) N/A	(6) N/A	DB	WB	-38.6	23.6	3.2	1.4	-40.9	24.6	-42.8	25.4	-44.5	26.2	-46.9	27.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.7	-15.5	-14.2	-7.3	1.1	10.0	15.8	18.6	17.1	13.0	5.5	-2.3	-10.8
	DBStd	13.33	8.34	7.35	7.58	5.71	5.12	4.10	3.29	3.40	4.61	4.85	5.80	7.24
	HDD10.0	3540	790	678	537	271	64	4	0	1	21	158	371	646
	HDD18.3	5821	1048	911	794	517	262	96	37	64	168	399	620	904
	CDD10.0	866	0	0	1	3	65	178	267	221	112	18	1	0
	CDD18.3	105	0	0	0	0	4	20	45	27	9	0	0	0
	CDH23.3	882	0	0	1	1	53	206	379	180	60	3	0	0
	CDH26.7	199	0	0	0	0	13	46	96	35	10	0	0	0
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(15) Precipitation	PrecAvg	816	46	37	38	56	75	83	95	88	97	85	64	51
	PrecMax	1025	114	74	75	105	120	144	158	147	160	143	120	87
	PrecMin	545	6	4	2	19	38	50	51	38	40	32	30	19
	PrecStd	95	23	21	19	24	21	23	30	29	26	29	21	19
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.8	4.5	14.1	20.2	28.3	30.5	31.7	29.6	27.8	22.1	13.2	5.2
		MCWB	3.4	3.0	8.7	11.8	17.6	21.2	20.9	20.9	20.8	16.6	11.0	4.9
	2%	DB	1.4	1.2	8.9	15.9	24.3	27.9	29.2	27.4	24.9	18.8	9.6	2.5
		MCWB	1.1	0.2	4.3	8.2	15.0	18.8	20.0	19.7	19.0	14.3	7.8	2.0
	5%	DB	-0.3	-1.0	6.1	12.9	21.8	25.8	27.2	25.3	22.9	15.8	7.4	0.8
		MCWB	-0.7	-2.0	2.9	7.2	13.9	17.5	19.1	18.7	17.6	12.7	6.0	0.4
	10%	DB	-2.4	-2.9	3.6	10.1	19.2	23.5	25.3	23.4	20.7	13.1	5.3	-0.4
		MCWB	-2.7	-3.6	1.2	5.7	12.5	16.6	18.2	18.0	17.2	10.9	4.3	-0.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	3.0	11.5	13.8	20.7	23.1	23.7	22.8	21.9	17.5	11.2	4.8
		MCDB	3.8	4.1	12.8	19.1	25.4	28.4	29.2	27.1	26.3	20.6	12.7	5.0
	2%	WB	1.2	0.3	5.6	10.0	17.6	21.2	21.9	21.1	20.4	15.1	8.2	1.8
		MCDB	1.4	1.3	8.2	13.6	21.1	24.8	25.8	24.8	23.3	18.2	9.0	2.1
	5%	WB	-0.6	-1.7	3.3	7.8	15.6	19.5	20.8	20.0	19.0	13.3	6.1	0.5
		MCDB	-0.3	-1.1	5.5	11.8	19.9	22.9	24.3	23.2	21.7	15.6	7.1	0.7
	10%	WB	-2.8	-3.6	1.3	5.9	13.4	18.0	19.6	19.0	17.5	11.1	4.2	-1.0
		MCDB	-2.4	-3.0	3.2	9.7	17.8	21.8	23.2	22.1	20.1	13.1	5.3	-0.7
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	12.9	13.5	11.4	11.7	11.0	10.6	9.8	9.6	7.3	6.7	8.9
		MCDBR	7.9	10.5	14.0	16.2	15.4	14.2	13.0	12.2	12.5	11.9	9.1	7.0
	5% WB	MCWBR	7.6	9.4	10.1	9.5	7.5	6.7	5.6	5.8	6.8	7.6	7.4	6.9
		MCDBR	7.9	9.9	11.9	14.0	12.1	11.5	10.7	10.0	10.3	10.6	8.4	7.1
		MCWBR	7.7	9.2	9.3	9.4	7.4	6.7	5.8	5.5	6.3	7.7	7.4	7.0
(40) Clear-Sky Solar Irradiance	taub		0.264	0.275	0.307	0.339	0.355	0.374	0.389	0.380	0.358	0.322	0.288	0.255
	taud		2.272	2.243	2.246	2.353	2.415	2.379	2.368	2.386	2.433	2.522	2.494	2.370
	Ebn at Noon		809	885	908	912	907	888	868	860	847	831	791	784
	Edh at Noon		74	95	113	114	111	116	116	109	95	74	61	59
(44) All-Sky Solar Radiation	RadAvg		1.26	2.33	3.78	4.68	5.23	5.65	5.46	4.58	3.45	2.02	1.07	0.85
	RadStd		0.20	0.33	0.39	0.41	0.44	0.36	0.46	0.30	0.30	0.27	0.12	0.11

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	-1.16	-1.40	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page